Speaker Identification System Using MFCC Feature Extraction and Random Forest Classifier

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***Abstract*—Speaker identification, the task of determining *who* is speaking rather than *what* is being said, underpins a wide range of applications including voice-based authentication, forensic audio analysis, speaker diarization and personalized voice assistants. While modern deep speaker embeddings such as x-vectors achieve state-of-the-art accuracy, they demand large corpora, GPU infrastructure and long training cycles that are impractical for embedded or resource-constrained deployments. This paper presents a lightweight, fully reproducible closed-set speaker identification pipeline that combines classical Mel-Frequency Cepstral Coefficient (MFCC) features with a Random Forest classifier. A corpus of $5{,}000$ spoken-digit recordings from 10 distinct speakers (500 utterances per speaker) is resampled to 16 kHz, length-normalized to exactly one second, and converted into a 40×40 MFCC matrix that is flattened into a $1{,}600$-dimensional feature vector. A Random Forest of 100 trees is trained on a stratified $80/20$ split. The system attains 99.50% validation accuracy, misclassifying only 5 of $1{,}000$ held-out utterances, with a macro-averaged F1-score of 0.99. Four of the ten speakers are recognized perfectly, and a detailed confusion analysis shows that all residual errors converge on a single acoustic attractor, Speaker_08, which absorbs three of the five errors and consequently exhibits the lowest precision (0.97). The results demonstrate that carefully normalized cepstral features paired with a shallow ensemble remain highly competitive for small, closed-set speaker populations while training in minutes on commodity CPU hardware.**

***Index Terms*—speaker identification, MFCC, Random Forest, audio classification, feature extraction, librosa, machine learning, biometrics**

I. Introduction

The human voice is a behavioural biometric that carries, in addition to linguistic content, a rich set of speaker-dependent cues arising from the physical geometry of the vocal tract, glottal excitation characteristics, and habitual articulatory patterns. *Speaker recognition* exploits these cues and is conventionally divided into two problems: *speaker verification*, a binary decision on a claimed identity, and *speaker identification*, a multi-class decision over an enrolled population [9]. This work addresses the latter in its *closed-set, text-independent* form: every test utterance is assumed to originate from one of ten enrolled speakers, and the system is not permitted to rely on knowing which word was spoken.

Research in this area has progressed through three broad generations. The first relied on generative statistical modelling, most notably the Gaussian Mixture Model with Universal Background Model (GMM-UBM) adaptation of Reynolds *et al.* [3]. The second introduced low-dimensional total-variability embeddings, the i-vector framework of Dehak *et al.* [4], which decoupled speaker and channel variability. The third and current generation replaces hand-designed statistics with discriminatively trained neural embeddings such as d-vectors [5] and x-vectors [6], trained on corpora of the scale of VoxCeleb [7], [8].

Deep embeddings are unquestionably superior when thousands of speakers, unconstrained recording channels and hours of audio are available. However, a large class of practical systems does not resemble that setting at all. A smart-home controller distinguishing among family members, a laboratory access-control terminal, or an attendance system for a classroom typically involves fewer than a few dozen enrolled speakers, must run on a CPU or microcontroller, and must be retrainable in seconds when a member joins or leaves. For these deployments the relevant question is not whether a deep model *can* be more accurate, but whether a classical pipeline is *accurate enough* at a fraction of the computational and data cost.

This paper answers that question empirically. The contributions are as follows.

1) We describe a complete, end-to-end and fully reproducible speaker identification pipeline built entirely from open-source components (`librosa` [11] and `scikit-learn` [12]), covering audio ingestion, duration normalization, MFCC extraction, feature serialization, model training and model persistence.
2) We show that aggressive temporal normalization to a fixed one-second window, combined with a fixed-size 40×40 cepstral matrix, produces a representation on which a shallow ensemble separates ten speakers almost perfectly.
3) We report 99.50% validation accuracy on $1{,}000$ unseen utterances and provide a full per-class classification

report and confusion matrix, rather than a single headline number.

4) We perform a directed error analysis showing that the five residual errors are not uniformly distributed but concentrate on one speaker, and we discuss the diagnostic value of this asymmetry.
5) We critically examine the gap between 100.00% training accuracy and 99.50% validation accuracy, and identify a session-overlap confound that limits how far the reported figure may be generalized.

The remainder of the paper is organized as follows. Section II reviews related work. Section III characterizes the corpus. Section IV details the methodology. Section V states the experimental configuration. Section VI presents and discusses the results. Section VII states the limitations, and Section VIII concludes.

II. Related Work

A. Cepstral Feature Representations

Mel-Frequency Cepstral Coefficients were introduced by Davis and Mermelstein [1] and have remained the dominant hand-crafted representation in speech processing for over four decades. Their strength lies in three successive operations: a mel-scaled filterbank that mimics the non-linear frequency resolution of the human cochlea, a logarithmic compression that converts multiplicative channel effects into additive offsets, and a Discrete Cosine Transform that decorrelates the resulting log-energies so that a diagonal-covariance or feature-independent classifier remains valid. Crucially for the present task, the low-order cepstral coefficients encode the spectral envelope, which is governed by vocal-tract resonances and is therefore strongly speaker-dependent.

B. Statistical and Embedding-Based Speaker Models

GMM-UBM systems [3] model each speaker as an adapted mixture over a speaker-independent background, and dominated the field until i-vectors [4] compressed an entire utterance into a single low-dimensional total-variability vector amenable to channel compensation. The transition to deep learning began with d-vectors [5], which averaged frame-level activations of a trained DNN, and matured with x-vectors [6], where a time-delay network with a statistics-pooling layer is trained explicitly to discriminate speakers. Comprehensive surveys of this trajectory are given by Hansen and Hasan [9] and by Bai and Zhang [10].

C. Shallow Classifiers on Cepstral Features

Parallel to the deep-learning line, a substantial body of work applies conventional classifiers directly to aggregated MFCC features. Random Forests [2] are particularly well suited to this regime: they are invariant to monotone feature scaling, tolerate the moderate redundancy of adjacent cepstral frames, provide an implicit feature-importance ranking, require essentially no hyperparameter tuning to reach a strong baseline, and train in seconds on a CPU. The present work sits squarely in this tradition, and its contribution is a careful, transparent characterization of exactly how far such a pipeline can be pushed on a small closed set.

D. Spoken-Digit Corpora

The dataset used here follows the organizational convention of AudioMNIST [13], in which each recording is a single spoken digit and file names encode the digit, the speaker identity and the repetition index. Because every speaker in such a corpus pronounces exactly the same ten lexical items, the linguistic content is perfectly balanced across classes; any discriminative signal the classifier exploits must therefore be attributable to speaker characteristics rather than to vocabulary bias. This property makes the corpus an unusually clean testbed for isolating speaker-dependent acoustic cues.

III. Dataset Description

The corpus consists of $5{,}000$ single-channel `.wav` recordings distributed over ten speakers, organized as ten directories `Speaker_01` through `Speaker_10`, each containing exactly 500 files. The directory index defines the ground-truth class label.

File names follow the pattern `{digit}_{speakerID}_{repetition}.wav`. Within each speaker directory the digit field takes each value in $\{0, 1, \ldots, 9\}$ exactly 50 times, so every speaker contributes $10 \times 50 = 500$ utterances and the class distribution is perfectly balanced. This balance is important: it removes the need for class re-weighting, makes accuracy a meaningful headline metric, and ensures that macro- and micro-averaged scores are directly comparable. Table I summarizes the corpus.

TABLE I
Dataset Composition and Audio Properties

Property	Value
Number of speakers (classes)	10
Utterances per speaker	500
Total utterances	5,000
Lexical content per speaker	digits 0–9
Repetitions per digit per speaker	50
Audio format	mono `.wav`, PCM
Resampled rate	16,000 Hz
Normalized duration	1.000 s (16,000 samples)
Class balance	perfectly uniform
Training utterances (80%)	4,000
Validation utterances (20%)	1,000

Raw file sizes range from approximately 39 kB to 79 kB, implying native durations that vary by roughly a factor of two across recordings. This variability is precisely what motivates the duration-normalization stage described in Section IV, since a fixed-length feature vector is a prerequisite for the chosen classifier.

A minor bookkeeping observation is worth recording for reproducibility: the embedded speaker identifiers in the `Speaker_06` and `Speaker_07` directories are transposed with respect to their directory names. Each directory is nevertheless internally consistent, containing recordings from one

and only one embedded identifier, so the labels supplied to the classifier remain clean and the reported results are unaffected. The transposition affects only the mapping back to original corpus identifiers.

IV. Methodology

The pipeline is deliberately partitioned into two independent stages: an offline feature-extraction stage that converts the audio corpus into a serialized numerical matrix, and a training stage that consumes that matrix. This separation means the expensive audio decoding is performed exactly once, after which any number of classifiers may be trained and compared in seconds.

A. Stage 1: Audio Ingestion and Duration Normalization

Each file is decoded and resampled to a common rate $f_s = 16{,}000$ Hz. This rate retains all spectral content below the 8 kHz Nyquist limit, which comfortably covers the first four formants and the fricative energy that carry the bulk of speaker-discriminative information, while halving the data volume relative to a 32 kHz representation.

Let $x[n]$ denote the decoded waveform of length L and let $T = 16{,}000$ be the target length corresponding to one second. The normalized signal $\tilde{x}[n]$ is obtained by truncation or zero-padding:

$$\tilde{x}[n] = \begin{cases} x[n], & 0 \le n < T, \quad \text{if } L \ge T \\ \begin{cases} x[n], & 0 \le n < L \\ 0, & L \le n < T \end{cases} & \text{if } L < T \end{cases} \tag{1}$$

Every utterance is thereby mapped to an identical support of exactly $16{,}000$ samples. Because the corpus consists of isolated digits whose articulation is typically well under one second, the dominant operation in practice is right-side zero-padding rather than truncation, so linguistic content is preserved rather than clipped.

B. Stage 2: MFCC Feature Extraction

The normalized waveform is transformed into the cepstral domain. The signal is decomposed into overlapping short-time frames using an FFT length of $2{,}048$ samples (128 ms) and a hop length of 512 samples (32 ms), yielding

$$N_{\text{frames}} = 1 + \left\lfloor \frac{T}{H} \right\rfloor = 1 + \left\lfloor \frac{16000}{512} \right\rfloor = 32 \tag{2}$$

frames per utterance under centered framing. The magnitude spectrum of each frame is passed through a mel-scaled triangular filterbank, where the perceptual mel scale is related to linear frequency f in hertz by

$$m(f) = 2595 \log_{10}\!\left(1 + \frac{f}{700}\right). \tag{3}$$

Denoting by E_k the energy in the k-th of M mel bands, the cepstral coefficients are obtained by applying a Type-II Discrete Cosine Transform to the log-compressed band energies:

$$c_i = \sum_{k=1}^{M} \log(E_k)\, \cos\!\left[\frac{i\left(k - \frac{1}{2}\right)\pi}{M}\right], \quad i = 1, \ldots, D \tag{4}$$

with $D = 40$ retained coefficients. The logarithm converts multiplicative gain and channel effects into additive terms, and the DCT concentrates the spectral-envelope information into the leading coefficients while approximately decorrelating them.

C. Stage 3: Fixed-Size Feature Vector Construction

Equations (2) and (4) yield a 40×32 matrix per utterance. To guarantee a constant input dimensionality irrespective of framing parameters, the time axis is further normalized to a fixed width $W = 40$. Let $\mathbf{C} \in \mathbb{R}^{D \times N_{\text{frames}}}$; then

$$\mathbf{C}' = \begin{cases} \left[\, \mathbf{C} \;\middle|\; \mathbf{0}_{D \times (W - N_{\text{frames}})} \,\right], & N_{\text{frames}} < W \\ \mathbf{C}[:,\, 1{:}W], & N_{\text{frames}} \ge W \end{cases} \tag{5}$$

Finally $\mathbf{C}'$ is flattened in row-major order into a single vector

$$\mathbf{v} = \operatorname{vec}(\mathbf{C}') \in \mathbb{R}^{1600}, \qquad 40 \times 40 = 1600. \tag{6}$$

Each of the $5{,}000$ utterances is thus represented by a $1{,}600$-dimensional vector, and the complete design matrix has shape 5000×1600. The stacked features $\mathbf{X}$ and integer labels $\mathbf{y}$ are serialized to a compressed `.npz` archive so that the training stage never re-reads the audio.

We note explicitly that since $N_{\text{frames}} = 32 < W = 40$, the final 8 columns of every feature matrix are structurally zero. These 320 constant dimensions carry no information; they are retained here for architectural generality but are, as discussed in Section VII, a source of avoidable representational overhead.

D. Stage 4: Random Forest Classification

The classifier is a Random Forest [2], an ensemble of $B = 100$ CART decision trees. Each tree h_b is grown on an independent bootstrap resample of the training set, and at every node a random subset of features is considered for splitting. Node purity is measured by the Gini index,

$$G(t) = 1 - \sum_{c=1}^{C} p_{c|t}^2, \tag{7}$$

where $p_{c|t}$ is the proportion of training samples of class c reaching node t and $C = 10$. The ensemble prediction is the majority vote

$$\hat{y}(\mathbf{v}) = \arg\max_{c \in \{1,\ldots,C\}} \sum_{b=1}^{B} \mathbb{I}[h_b(\mathbf{v}) = c]\,. \tag{8}$$

The double randomization over samples and features decorrelates the individual trees, so that averaging reduces variance without materially increasing bias. Two properties make this choice particularly appropriate here. First, decision trees are

invariant to monotone transformations of individual features, so the widely differing dynamic ranges of low- and high-order cepstral coefficients require no standardization. Second, the ensemble is insensitive to the 320 structurally-zero dimensions identified above, since a constant feature can never reduce the Gini impurity and is therefore never selected as a split.

E. Algorithmic Summary

Algorithm 1 MFCC–Random Forest Speaker Identification

Require: Root directory $\mathcal{D}$ with subfolders `Speaker_01..10`
Ensure: Trained model $\mathcal{M}$, evaluation report $\mathcal{R}$

$\mathbf{X} \leftarrow [\,], \quad \mathbf{y} \leftarrow [\,]$
for $\ell \leftarrow 0$ **to** 9 **do**
  for each file f in `Speaker_`$(\ell{+}1)$ **do**
    $x \leftarrow \textsc{Load}(f,\ f_s = 16000)$
    $\tilde{x} \leftarrow \textsc{FixLength}(x,\ T = 16000)$ ▷ Eq. (1)
    $\mathbf{C} \leftarrow \mathrm{MFCC}(\tilde{x},\ D = 40)$ ▷ Eq. (4)
    $\mathbf{C}' \leftarrow \textsc{PadOrCrop}(\mathbf{C},\ W = 40)$ ▷ Eq. (5)
    append $\mathrm{vec}(\mathbf{C}')$ to $\mathbf{X}$; append ℓ to $\mathbf{y}$
  end for
end for
$\textsc{Serialize}(\mathbf{X}, \mathbf{y}) \rightarrow$ `processed_data.npz`
$(\mathbf{X}_{tr}, \mathbf{X}_{te}, \mathbf{y}_{tr}, \mathbf{y}_{te}) \leftarrow \textsc{StratifiedSplit}(\mathbf{X}, \mathbf{y}, 0.2)$
$\mathcal{M} \leftarrow \textsc{RandomForest}(B = 100).\textsc{Fit}(\mathbf{X}_{tr}, \mathbf{y}_{tr})$
$\hat{\mathbf{y}} \leftarrow \mathcal{M}.\textsc{Predict}(\mathbf{X}_{te})$
$\mathcal{R} \leftarrow \textsc{Evaluate}(\mathbf{y}_{te}, \hat{\mathbf{y}})$ ▷ Eq. (8)
$\textsc{Persist}(\mathcal{M}) \rightarrow$ `speaker_identification_model.pkl`
return $\mathcal{M}, \mathcal{R}$

V. Experimental Setup

The corpus is partitioned into training and validation sets in an $80/20$ ratio using *stratified* sampling, which preserves the exact 10% class proportion in both partitions and guarantees exactly 400 training and 100 validation utterances per speaker. A fixed random seed of 42 governs both the split and the forest construction, making every reported number bit-for-bit reproducible. The complete configuration is given in Table II.

Performance is reported using accuracy, together with per-class precision, recall and F1-score. For class c, with TP_c, FP_c and FN_c denoting true positives, false positives and false negatives:

$$P_c = \frac{TP_c}{TP_c + FP_c}, \qquad R_c = \frac{TP_c}{TP_c + FN_c}, \tag{9}$$

$$F1_c = 2 \cdot \frac{P_c \cdot R_c}{P_c + R_c}. \tag{10}$$

Because the validation set is exactly balanced, the macro average (unweighted mean over classes) and the weighted average coincide, and both are directly comparable to overall accuracy.

TABLE II
Pipeline and Model Configuration

Parameter	Setting
Feature extraction	
Sampling rate f_s	16,000 Hz
Resampling method	`kaiser_fast`
Target duration T	16,000 samples (1 s)
FFT window	2,048 samples (128 ms)
Hop length H	512 samples (32 ms)
Frames per utterance	32
MFCC coefficients D	40
Padded time width W	40
Feature vector dimension	1,600
Classifier	
Algorithm	Random Forest
Number of estimators B	100
Split criterion	Gini impurity
Maximum depth	unbounded
Random seed	42
Protocol	
Train / validation split	80% / 20%, stratified
Training samples	4,000
Validation samples	1,000
Design matrix shape	5000×1600
Environment	
Language	Python 3
Audio processing	`librosa` [11]
Machine learning	`scikit-learn` [12]
Numerics	`NumPy`
Model persistence	`joblib`
Hardware	commodity CPU (no GPU)

VI. Results and Discussion

A. Overall Performance

The trained model classifies 995 of the $1{,}000$ held-out utterances correctly, for a validation accuracy of 99.50%. Training accuracy reaches 100.00%. Headline results are collected in Table III.

TABLE III
Overall Classification Performance

Metric	Value
Training accuracy	100.00%
Validation accuracy	**99.50%**
Correctly classified (validation)	995 / 1,000
Misclassified (validation)	5 / 1,000
Macro-averaged precision	0.99
Macro-averaged recall	0.99
Macro-averaged F1-score	0.99
Perfectly classified speakers	4 / 10
Error rate	0.50%

B. Per-Class Results

Table IV gives the full classification report. Speakers 01 through 04 are recognized without a single error in either direction, achieving precision, recall and F1 all equal to 1.00. The remaining six speakers deviate from perfection by at most one or two misassignments.

TABLE IV
PER-CLASS CLASSIFICATION REPORT ON THE VALIDATION SET

Speaker	Precision	Recall	F1-score	Support
Speaker_01	1.00	1.00	1.00	100
Speaker_02	1.00	1.00	1.00	100
Speaker_03	1.00	1.00	1.00	100
Speaker_04	1.00	1.00	1.00	100
Speaker_05	1.00	0.99	0.99	100
Speaker_06	1.00	0.99	0.99	100
Speaker_07	0.99	1.00	1.00	100
Speaker_08	**0.97**	0.99	0.98	100
Speaker_09	1.00	0.99	0.99	100
Speaker_10	0.99	0.99	0.99	100
Accuracy		**0.99**		**1000**
Macro avg	1.00	0.99	1.00	1000
Weighted avg	1.00	0.99	1.00	1000

C. Confusion Matrix

Table V presents the 10×10 confusion matrix, where rows are true speakers and columns are predicted speakers. The matrix is almost perfectly diagonal, with exactly five off-diagonal entries, each of magnitude one.

TABLE V
CONFUSION MATRIX (ROWS: TRUE CLASS, COLUMNS: PREDICTED CLASS)

	S01	S02	S03	S04	S05	S06	S07	S08	S09	S10
S01	**100**	0	0	0	0	0	0	0	0	0
S02	0	**100**	0	0	0	0	0	0	0	0
S03	0	0	**100**	0	0	0	0	0	0	0
S04	0	0	0	**100**	0	0	0	0	0	0
S05	0	0	0	0	**99**	0	0	0	0	1
S06	0	0	0	0	0	**99**	0	1	0	0
S07	0	0	0	0	0	0	**100**	0	0	0
S08	0	0	0	0	0	0	1	**99**	0	0
S09	0	0	0	0	0	0	0	1	**99**	0
S10	0	0	0	0	0	0	0	1	0	**99**

Figure 1 reproduces the raw console output from which Tables III–V are derived.

D. Error Analysis

The five errors are enumerated in Table VI. Their distribution is markedly non-uniform and is the most informative aspect of the result.

TABLE VI
ENUMERATION OF THE FIVE VALIDATION ERRORS

#	True speaker	→	Predicted speaker
1	Speaker_05	→	Speaker_10
2	Speaker_06	→	Speaker_08
3	Speaker_08	→	Speaker_07
4	Speaker_09	→	Speaker_08
5	Speaker_10	→	Speaker_08

Three observations follow.

First, Speaker_08 acts as an acoustic attractor. It receives three of the five false assignments, drawn from three different source speakers (06, 09 and 10). Because precision penalizes

```
PS D:\CSE445 Project\speech-recognition-10-speakers> python model_training.py
[INFO] Loading preprocessed MFCC matrices from dataset_processor.py...
[INFO] Loaded 5000 samples with feature dimensions: 1600
[INFO] Constructing Voice Identification Model (Random Forest)...
[INFO] Training Model...
[SUCCESS] Training Complete.
 -> Training Accuracy: 100.00%
 -> Validation Accuracy: 99.50%

[EVALUATION REPORT] Classification Detail:
              precision    recall  f1-score   support

  Speaker_01       1.00      1.00      1.00       100
  Speaker_02       1.00      1.00      1.00       100
  Speaker_03       1.00      1.00      1.00       100
  Speaker_04       1.00      1.00      1.00       100
  Speaker_05       1.00      0.99      0.99       100
  Speaker_06       1.00      0.99      0.99       100
  Speaker_07       0.99      1.00      1.00       100
  Speaker_08       0.97      0.99      0.98       100
  Speaker_09       1.00      0.99      0.99       100
  Speaker_10       0.99      0.99      0.99       100

    accuracy                           0.99      1000
   macro avg       1.00      0.99      1.00      1000
weighted avg       1.00      0.99      1.00      1000


[EVALUATION REPORT] Confusion Matrix Generated:
[[100   0   0   0   0   0   0   0   0   0]
 [  0 100   0   0   0   0   0   0   0   0]
 [  0   0 100   0   0   0   0   0   0   0]
 [  0   0   0 100   0   0   0   0   0   0]
 [  0   0   0   0  99   0   0   0   0   1]
 [  0   0   0   0   0  99   0   1   0   0]
 [  0   0   0   0   0   0 100   0   0   0]
 [  0   0   0   0   0   0   1  99   0   0]
 [  0   0   0   0   0   0   0   1  99   0]
 [  0   0   0   0   0   0   0   1   0  99]]

[SAVED] Model saved successfully as speaker_identification_model.pkl
PS D:\CSE445 Project\speech-recognition-10-speakers>
```

Fig. 1. Console output of the training and evaluation run, showing the reported training and validation accuracy, the per-class classification report, the confusion matrix, and successful serialization of the trained model.

exactly these incoming errors, Speaker_08 records the lowest precision in the system at $0.97 = 99/102$, while its recall remains high at 0.99. The asymmetry between its precision and recall is diagnostic: the model is not failing to *find* Speaker_08, it is over-*assigning* to Speaker_08. This is the signature of a class whose learned region in feature space is slightly too permissive, typically because its cepstral envelope occupies a central, unremarkable position relative to its neighbours and therefore lies closest to several decision boundaries at once.

Second, no error is bidirectional except the 07/08 pair. Speaker_08 loses one utterance to Speaker_07 while Speaker_07 loses none in return, and Speaker_07's precision of $0.99 = 100/101$ reflects that single intrusion. A genuinely confusable speaker pair would normally produce symmetric leakage in both directions; the near-absence of symmetry here indicates that the errors are better explained by individual borderline utterances, most plausibly recordings degraded by low energy, clipping or truncation, than by an intrinsic similarity between two voices.

Third, all confusion is confined to the upper half of the speaker index. Speakers 01–04 are entirely error-free and Speakers 05–10 account for every mistake. Errors therefore cluster within one contiguous subgroup rather than spreading uniformly, which suggests a shared recording characteristic within that subgroup rather than ten independent failure

modes.

E. Discussion

The 0.50% gap between 100.00% training accuracy and 99.50% validation accuracy warrants comment. Fully grown, unpruned Random Forests are expected to interpolate their training data exactly, so perfect training accuracy is a structural property of the model class rather than evidence of pathological overfitting. The relevant evidence is instead the size of the generalization gap, and half a percentage point is small enough to indicate that the ensemble has captured genuinely transferable structure rather than memorized noise.

Two design decisions appear to be the principal contributors to the result. The first is the strict duration normalization of Equation (1). By forcing every utterance onto an identical one-second support, the pipeline eliminates duration as a nuisance variable, so that a given position in the flattened vector of Equation (6) corresponds to a consistent point in the temporal envelope across all $5{,}000$ recordings. Individual coordinates thereby become directly comparable across samples, which is exactly the condition under which axis-aligned decision-tree splits are most effective. The second is the balanced lexical content of the corpus. Since every speaker utters the identical set of ten digits an identical number of times, the classifier cannot achieve separation by exploiting vocabulary differences; whatever it learns must reside in speaker-dependent spectral-envelope characteristics.

Finally, the practical economics deserve emphasis. The forest comprises 100 trees over $1{,}600$ features, trains in a matter of seconds on a CPU without any GPU involvement, and serializes to a single portable artifact. For a closed set of ten enrolled speakers, this is an entirely favourable trade against a deep embedding extractor that would require orders of magnitude more data, compute and engineering effort to reach a comparable operating point.

VII. Limitations

Intellectual honesty requires that the 99.50% figure be qualified rather than presented as a general claim about speaker identification.

Session and channel overlap. The train/test split is performed at the level of individual utterances, not recording sessions. Since all 500 recordings of a speaker were most plausibly captured in a small number of sessions using the same microphone and acoustic environment, training and validation utterances from the same speaker share channel characteristics. The classifier may therefore be partially identifying the *recording condition* rather than the *voice*. A session-disjoint or channel-disjoint protocol would yield a lower and more honest estimate of true speaker-discriminative power, and would be the single most valuable methodological improvement to this study.

Closed-set assumption. The system always emits one of ten labels and has no mechanism for rejecting an unenrolled speaker. Practical deployment would require an open-set formulation with a calibrated score threshold or an explicit out-of-set class.

Constrained lexical domain. All utterances are isolated spoken digits of approximately one second. Performance on continuous, conversational or spontaneous speech, where coarticulation and prosodic variation are far greater, is not established by these experiments.

Clean-condition evaluation. No additive noise, reverberation, codec compression or far-field capture was introduced. MFCCs are known to degrade substantially under such mismatch, and no robustness margin has been measured here.

Representational inefficiency. As established in Section IV, 8 of the 40 time columns are structurally zero, so 320 of the $1{,}600$ dimensions are constant and carry no information. Aligning W with the true frame count of 32 would reduce the feature dimension by 20% at no cost in accuracy.

Discarded temporal dynamics. Flattening the cepstral matrix into a vector destroys the explicit two-dimensional time–frequency adjacency structure, and no delta or delta-delta coefficients are computed. The velocity and acceleration of the spectral envelope, well established as strongly speaker-dependent, are therefore unavailable to the classifier.

Single-split evaluation. Results derive from one stratified hold-out split under a fixed seed. With only five errors in total, the confidence interval on the error rate is wide, and k-fold cross-validation with reported variance would give a substantially more trustworthy estimate.

Population scale. Ten speakers is a small closed set. Confusion probability grows with population size, and these figures should not be extrapolated to hundreds or thousands of enrolled identities.

VIII. Conclusion and Future Work

This paper presented a complete and reproducible speaker identification system combining 40-dimensional MFCC features with a 100-tree Random Forest classifier. Evaluated on $5{,}000$ utterances from 10 speakers under a stratified $80/20$ split, the system achieved 99.50% validation accuracy with a macro-averaged F1-score of 0.99, misclassifying only 5 of $1{,}000$ held-out recordings. Four speakers were identified without any error. A directed error analysis revealed that the residual errors are not randomly scattered but concentrate on a single acoustic attractor class, Speaker_08, whose precision–recall asymmetry (0.97 versus 0.99) precisely characterizes the failure mode as over-assignment rather than under-detection.

The broader conclusion is that for small, closed-set speaker populations recorded under consistent conditions, a classical cepstral pipeline paired with a shallow ensemble remains a highly effective and computationally economical solution, training in seconds on a CPU and deploying as a single portable artifact.

Future work will proceed along five directions. *(i)* Re-evaluation under a session-disjoint protocol, together with k-fold cross-validation and reported variance, to obtain a defensible estimate of true generalization. *(ii)* Augmentation of the feature set with delta and delta-delta cepstra, chroma, spectral contrast and zero-crossing statistics to restore the

temporal-dynamic information currently discarded by flattening. *(iii)* Comparative benchmarking against gradient-boosted trees [14], support vector machines and a convolutional network operating on the MFCC matrix as a two-dimensional image, which would preserve the time–frequency adjacency that flattening destroys. *(iv)* Robustness evaluation under additive noise, reverberation and codec compression, together with an open-set extension using a calibrated rejection threshold for unenrolled speakers. *(v)* Deployment of the serialized model behind a real-time inference interface for live microphone input, and scaling of the enrolled population to assess how the error rate grows with the number of identities.

Code and Data Availability

The complete implementation, including the feature-extraction script, the training and evaluation script, the dependency specification and the full audio corpus, is publicly available at:

https://github.com/Ashfaq127/speech-recognition-10-speakers

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